

Explain/prove everything.

You may use std facts about row reduction to solve linear systems + matrix multiplication

① Write solution set to $Ax=0$ and find a basis for it

a) $A = \begin{bmatrix} 1 & a & 0 & p \\ 0 & 1 & 0 & q \\ 0 & 0 & b & r \end{bmatrix}$ $a, b, p, q, r \in \mathbb{R}$
 $b \neq 0$

b) Redo when $b=0$

③ Let A be an $r \times c$ matrix (a) show (as directly as you can):

columns of A are independent $\iff$ rows of A span $\mathbb{R}^c$

(b) Deduce rows of A are indep $\iff$ cols of A span $\mathbb{R}^r$.

② $A \xrightarrow[\text{operations}]{\text{row}} B$ True or false?



- row space(A) = row space(B)
- rows of A are lin indep $\Leftrightarrow$ rows of B are lin indep
- col space(A) = col space(B)
- cols of A lin indep $\Leftrightarrow$ cols of B lin indep

③ let $v \in \mathbb{R}^3, v \neq 0$.

Show that I can find two linear equations in three variables whose solution set is the line spanned by v (i.e. the set $\{av \mid a \in \mathbb{R}\}$)

⑤ Let $f: V \rightarrow W$ be a bijective linear map. Show that the inverse map is also linear.



Exercise in language